

The barriers to environmental sustainability in post-disaster settings: a case study of transitional shelter implementation in Haiti

Daniel Abrahams Doctoral Candidate, Department of Geography, University of South Carolina, South Carolina, United States¹

Disaster recovery operations that do not account for environmental sustainability (ES) risk exacerbating the impact of the disaster and hindering long-term recovery efforts. Yet aid agencies do not always consider ES. This research is a case study of the recovery that followed the 2010 earthquake in Haiti. Using timber and concrete procurement as proxies for broader post-disaster operations, research examined perceptions of ES as well as attempts at and barriers to incorporating it into programming. Identified barriers can be grouped into two categories: (1) prioritisations and perceptions within the disaster response sector that resulted in limited enthusiasm for incorporating ES into programming, and (2) structural and organisational barriers within the disaster response framework that impeded ES attempts and served as a further disincentive to incorporating ES into programming. As a result of those barriers, incorporation of ES was sporadic and inconsistent and often depended on the capacity and motivation of specific implementers.

Keywords: disaster response, disaster recovery, disaster resilience, earthquake, environmental sustainability, Haiti, humanitarian assistance, procurement, transitional shelter

Introduction

Of the four stages of a disaster—mitigation, preparedness, response and recovery—recovery is the least studied (Berke and Beatley, 1997; Ganapati and Ganapati, 2009; Berke, Kartez and Wenger, 1993; Peacock, Dash and Zhang, 2007). Consequently, there is very limited research on the challenges, barriers and strategies to incorporating environmental sustainability (ES) into post-disaster recovery.

Expanding global population and increasing rates of urbanisation will result in greater population density in hazard-prone lands, rendering more people and property vulnerable to disasters (IPCC, 2008; Mainka and McNeely, 2011; O'Brien et al., 2006).² Moreover, due to the increasing severity of the effects of climate change, the likelihood of high-impact natural hazards will continue to increase (Mainka and McNeely, 2011; Pelling, 2007; van Aalst, 2006). Thus, it is likely that the rate and severity of disasters will continue to increase—and that the demand for post-disaster recovery efforts will intensify accordingly.

To ensure that poor resource management does not compound future disasters, it is paramount that both the post-disaster and conservation communities actively work

to understand the barriers to ES in a post-disaster setting. Post-disaster operations that do not incorporate ES risk exacerbating the impact of a disaster and hindering long-term recovery efforts, and thereby disaster resilience. Yet organisations do not always consider ES during post-disaster response and recovery and those that do often encounter logistical, organisational and structural obstacles.

The following analysis is based on a case study of transitional shelter (t-shelter) implementation in Haiti following the 12 January 2010 earthquake; it utilises timber and concrete procurement as proxies to examine broader post-disaster issues. Data collection and analysis focused on localised environmental issues that can inhibit long-term recovery, create new risks or exacerbate old ones. This research reveals specific barriers that impeded the incorporation of ES during t-shelter implementation in the Haiti recovery process; the results are intended to be transferable to other post-disaster settings.

Background

Effective post-disaster shelter programming is central to the success of post-disaster response and recovery. A slow housing recovery can impede other aspects of the response, including public health and socioeconomic sustainability (Ashmore et al., 2003; Davis, 1978; Leon et al., 2009; Peacock, Dash and Zhang, 2007). Quarentelli (1982) describes four post-disaster shelter stages that overlap or, in some cases, take place simultaneously: emergency shelter, temporary shelter, temporary housing and permanent housing. In contrast to shelter, housing encompasses the non-physical aspects of shelter, such as physical and mental health, security and household responsibilities (Quarentelli, 1982). In addition, temporary shelter can transition to permanent housing should the displaced be unable or unwilling to return to permanent homes (Berke and Beatley, 1997; Peacock, Dash and Zhang, 2007; Quarentelli, 1982). Under Quarentelli's framework, t-shelters fit most closely with temporary housing.

T-shelters can be loosely defined as 'more than a tent, less than a house'. Stated more formally, t-shelter 'provides a habitable covered living space and a secure, healthy living environment, with privacy and dignity, to those within it, during the period between a conflict or natural disaster and the achievement of a durable shelter solution' (Corsellis and Vitale, 2005, p. 11). T-shelters are generally constructed during the short- to mid-term recovery phase between immediate life-saving activities and the reconstruction and reestablishment of permanent homes and communities, which can take years (Alexander, 2004; Leon et al., 2009).

Some of the most vexing challenges of post-disaster response and recovery are embodied by post-disaster shelter operations. For example, the urgency of post-disaster activities can amplify the weaknesses that originally made an area vulnerable to natural hazards.³ An inherent challenge for the post-disaster community, particularly in terms of shelter, is the need and desire to rebuild quickly, and the simultaneous need to rebuild safely and equitably (Kates et al., 2007; Nelson, Ehrenfeucht and Laska, 2007; Olshansky, 2006). Speed is important not only because of the physical

properties of shelter, but also because it helps drive the economic facets of recovery, such as through employment for the displaced (Olshansky, 2006). In addition, if responding agencies are not fast enough, displaced households will start to rebuild where they deem appropriate, including unsafe areas, thereby reproducing vulnerabilities that existed before the impact (Nelson, Ehrenfeucht and Laska, 2007; Olshansky, 2006).

The logistical components of reconstruction, including transportation, warehousing, purchasing and distribution, are rife with complications (Zuo et al., 2008). Following large-scale disasters, the imbalance caused by a lack of available reconstruction supplies and extremely high demand typically results in a sharp escalation in construction costs (Chang et al., 2010; GRRT, 2010; Ingram et al., 2006). This excess demand can result in unsustainable procurement, which can increase vulnerability to future disasters or further exacerbate the current disaster.⁴ For example, in Aceh, following the 2004 Boxing Day tsunami, the resulting need to import key construction materials led to as much as a 180% increase in the cost of construction materials, thereby unintentionally encouraging rapid—potentially illegal—deforestation (Chang et al., 2010; Nazara and Resosudarmo, 2007, p. 40). Uncontrolled deforestation as a result of increased demand for timber for reconstruction can exacerbate vulnerability via silting, flooding, landslides and water contamination.⁵

Integrating the principles of ES into post-disaster response and recovery can encourage and support sustainable development, disaster resilience and long-term reconstruction of the affected region. The recovery period provides a unique opportunity to mitigate long-term risk from hazards and increase disaster resilience. A recovery that does not incorporate ES could hinder long-term socioeconomic goals and risks exacerbating the damage of the disaster.⁶

Incorporating environmental concerns supports the oft-referenced goal of ‘building back better’ and the less referenced, but potentially more appropriate goal of ‘building back safer’ (Berke and Beatley, 1997; Kennedy et al., 2008; Smith and Wenger, 2007). Indeed, the term ‘safer’ eliminates some of the ambiguities of the term ‘better’, which can mean ‘more aesthetically pleasing’ without implying ‘safer’, for example (Kennedy et al., 2008). ‘Safer’ also incorporates the importance of more sustainable land and resource usage.

What is often deemed a ‘natural disaster’ is really a triggering event releasing pressure that existed due to pre-existing social issues (Lizarralde, Johnson and Davidson, 2009; Wisner et al., 2004). Disasters tend to accelerate socioeconomic trends (Kates et al., 2007). In the developing world, social, economic and environmental challenges have left communities dangerously and disproportionately vulnerable to natural disasters (Aquilino, 2011; O’Brien et al., 2006; van Breda and Laprade, 2008; Roseberry, 2008). Those risks are not just physical; losses due to natural hazards are twenty times greater in developing countries than they are in the developed world (Barakat, 2003).

According to Guarnacci, ‘what distinguishes sustainable reconstruction from conventional is a focus on long-term community resilience, in order to “build back better” and promote a culture of prevention’ (Guarnacci, 2012, p. 82). The current research builds on past definitions of sustainable development⁷ and applies a modified definition of environmental sustainability created specifically for the post-disaster context:

Sustainable post-disaster activities provide resources to affected citizens to ensure health and safety and to promote redevelopment, without causing further damage to land or existing structures, exacerbating the impacts of the disaster, or placing undue stress on the natural environment.

This research, however, did not concentrate on the semantics of sustainability; rather, the focus was on the core principles of ES, such as minimising natural resource usage and total wastage.

Haiti background information

Owing to its socioeconomic and governmental challenges, political instability and hazard-prone geography, Haiti is one of the most disaster-prone countries in the world. When the earthquake struck, the country was still recovering from four hurricanes that killed more than 800 people and caused more than \$900 million in damage in 2008; in 2004 Hurricanes Jeanne and Ivan killed more than 14,000 people (FAO, 2007).

Prior to the earthquake, 70% of Haiti's government budget was provided by foreign aid; moreover, there was a lack of modern building codes and the codes that did exist were rarely enforced (ICG, 2010, p. 2). Haiti also suffers from immense environmental degradation; for example, prior to the earthquake, less than 1% of its original forest cover remained and 58% of the population lacked access to clean water (WWF and IASC, 2010; OSE, n.d.). Further exacerbating the challenges of the recovery, the bulk of the destruction took place in a densely populated, extremely poor urban area. In 2009, before the earthquake, the annual income per capita in Haiti was \$650, or \$1.78/day (UNEP, 2011, p. 4). Port-au-Prince has a population density of 28,757 people/km² (Joseph and Wang, 2010, p. 128). As a point of comparison, Manhattan has a population density of 26,924 people/km² (USCB, n.d.). As a result of the earthquake:

- more than 220,000 people died (UNEP, 2011, p. 10);
- there was \$7.8 billion in damages, or 121% of Haiti's 2009 GDP (UNEP, 2011, p. 16);
- 1.5 million people moved into camps and another 600,000 moved to another area of the country, meaning that more than 20% of Haiti's population left their homes (UNEP, 2011, p. 10);
- roughly 110,000 t-shelters were built (IASC E-Shelter and CCCM Cluster, 2012a, p. 1); and
- roughly 10 million m³ of rubble was left in its wake (USAID, 2012, p. 1).

Research aims and methods

This analysis is based on a case study of t-shelter implementation in Haiti following the 12 January 2010 earthquake. Informants consistently described the recovery efforts as among the most difficult, if not the most difficult of their careers. Extreme or

atypical cases, which tend to activate more actors and mechanisms, often reveal more information and therefore are the best case studies (Flyvbjerg, 2006).

This study aims to (1) address gaps in the existing knowledge of post-disaster environmental sustainability efforts, and (2) examine the benefits, trade-offs and practicality of considering ES in a post-disaster setting. The research is focused on localised environmental issues stemming from post-disaster activities that can inhibit long-term recovery, create new risks or exacerbate old ones, such as increased vulnerability to landslides due to unsustainable resource extraction. The research examined timber and concrete used in t-shelters as proxies for broader post-disaster operations. The primary data collection consisted of semi-structured interviews with 37 informants and direct observation during a field visit to Haiti in May 2012.⁸

T-shelter implementation was chosen because it is representative of the broader challenges associated with incorporating ES into disaster response. T-shelter implementation involves procurement and complex supply chains, multiple stakeholders and significant material inputs. In addition, land use, land tenure, institutional planning, urban and rural issues, community involvement, budgeting, planning and speed in a post-disaster recovery were all critical factors in t-shelter programming.⁹ Informants provided insight and details on t-shelters based on their recovery experiences. Examining t-shelter implementation in Haiti was ideal in terms of timing: the research started in August 2011, at the time many organisations were winding down, or had just finished, their t-shelter operations. Thus, informants had sufficient experience, yet maintained strong levels of institutional memory.

Semi-structured interviews were the primary source of data and covered a wide range of topics, including procurement methods, implementation strategies, perceptions of ES, attempts and challenges to incorporating ES into programming and general decision paths. Informants' perceptions of specific activities and actions varied.

Table 1. Informant details

Informant type	Definition of informant type	Total interviewed
Implementer	Implementers were on the ground, either implementing t-shelter programmes or performing shelter-related activities. This group consisted predominantly of people who worked with international non-governmental organisations (NGOs).	19
Organiser	Organisers worked for organisations that tried to ensure that all parties were 'on the same page'. The vast majority of these informants worked for the shelter cluster. ¹⁰ In Haiti, nearly all major actors participated in the cluster system.	7
Donor	Donors worked for the organisations that provided funding for implementers' activities. These tended to be governmental or bilateral organisations.	6
Expert	Experts represented a wide range of actors with significant experience in disaster response and recovery, shelter, energy, rubble or timber procurement. Some worked in Haiti, others did not. They provided data on specific issues, helped confirm data and provided context and guidance on technical issues.	5

Source: author.

Multiple informants were thus asked about the same experiences, particularly if multiple types of parties—such as donors and implementers—were engaged in the same activity. Table 1 provides details on the informants.

Informants were identified via professional networking, referrals from other informants and research, including looking up individuals listed on organisations' websites. Interviewed participants represented 20 agencies; non-agency-affiliated consultants were classified as individual agencies. Multiple informants were 'hybrids'—for example, experts who worked as consultants to donors or implementers. To ensure confidentiality, informants were given only one title. In addition, some quotes have been modified to ensure confidentiality. The author employed the following methods to inform the analysis:

- *Interviews*: Interviews started with general topics and then relied on extensive probing and more directed questions based on issues brought up by informants (Schoenberger, 1991). Interviews focused on t-shelters but often branched into other areas owing to the interdisciplinary nature of disasters; for example, rubble emerged as a critical issue.
- *Direct observation*: The field research consisted of visits to multiple sites with two leading implementers. The researcher undertook direct observation (both non-intrusive and participant), interviewed key actors on site and was able to compare and contrast different types of transitional shelters (Bernard, 2006). Time on the ground was critical in gaining a contextual understanding of the challenges of working in Haiti and in understanding the inputs of t-shelters as well as the complexities related to rubble management.
- *Review of secondary data*: The review involved analysis of grey data and of reports posted on sheltercentre.org, shelterhaiti.org, reliefweb.int, and the websites of informants' organisations, as well as of internal documents provided directly by informants (including data sets and non-confidential reports).
- *Analysis*: Iterative data analysis was used to refine research questions and inform data collection. Issues were explored until saturation, that is, until no new categories or themes emerged (Corbin and Strauss, 2008). Data not centred on informants' direct experience was confirmed via published documents and/or informants with direct knowledge of the topic. To ensure accuracy, the vast majority of interviews were recorded and then transcribed verbatim by the author. Data was then examined to identify patterns and disconnects among informants, general decision paths and potential intervention points to increase ES. Data collected during interviews was triangulated with direct observation and secondary sources.

Of note are the on-the-ground realities of working in such an emotionally charged setting. Informants spoke passionately not only about their jobs, but also about the horrors that befell an already beleaguered country. In addition to the logistical, political and physical challenges, the emotional challenges were also considered in data collection and analysis. As put by Informant 7, an organiser, regarding working in Haiti after the earthquake: 'It was difficult. It was busy. It was crazy. It was a nightmare.'

Transitional shelter implementation: assessing processes

T-shelters in Haiti generally consisted of a foundation or concrete slab floor (which served as a risk reduction tool against the high winds of the hurricane season), a wooden frame, plywood or tarp siding and a corrugated galvanised iron roof.¹¹ Although t-shelters are traditionally designed to last 2–5 years, many informants predicted that the structures would serve as beneficiaries' houses for the next 10–15 years. There was no expected lifespan of the material inputs of t-shelters. The shelter cluster stipulated that shelters should last at least three years and have at least 18 m² of living space (IASC Haiti Shelter Cluster, 2010, p. 10).

The shelters were intended to cost \$1,000–\$1,500, including transport and labour, excluding taxes (IASC Haiti Shelter Cluster, 2010, p. 10). Informants noted that actual costs far exceeded that target. The excess costs were driven by unexpected expenses related to security, shipping, storage, customs and other logistical aspects of operations, such as buying and transporting clean water for workers. Figure 1 provides an example of a t-shelter.

Material inputs more representative of permanent (as compared to transitional) shelters also contributed to the higher-than-intended cost of t-shelters. Despite the relatively high permanence of material inputs, many shelters observed during field visits—particularly those with tarp siding—were starting to show signs of weathering.

Informants attributed the increased permanence of material inputs to a 'lottery effect' and more pronounced risk aversion. A lottery effect is when Community A,

Figure 1. A plywood-sided transitional shelter



Note: This was described as, and appeared to be, of higher permanence and quality than the majority of t-shelters.

Source: author.

served by organisation X, receives top-of-the-line shelters, while neighbouring Community B, served by organisation Y, receives more basic shelters. Community B then insists on t-shelters of comparable quality to Community A. This was especially pervasive in Haiti due to the influx of money and inexperienced, non-traditional actors who were not familiar with the traditional post-disaster framework and did not always coordinate with other agencies. Informants felt that despite the technical and complex nature of shelter programming (including a need for coordination), non-experienced actors would gravitate towards t-shelter construction owing to the obvious need and tangible outputs.

In addition, a heightened sense of risk aversion also increased the cost and permanence of t-shelters. As noted by Informant 29, a donor with an environmental background:

Many of the organisations were so scared of building a t-shelter [incapable of withstanding hurricanes]. And so this is a big question that everybody had: how are we going to respond to a Category 4 hurricane? [. . .] Look at the foundations, the foundations [weigh] more than a ton. To have this sort of groundings, you have to have pylons, then you have to have various mechanisms for holding the roof, all of those things. Then you see the cost of the t-shelter is \$3,000 to \$4,000.

T-shelters are intended to serve as a step between emergency shelters and permanent housing. In Haiti, there was a massive effort to distribute emergency shelter (tents or makeshift tent-like structures, often referred to as *anba prelas*). By July 2010 more than 1.5 million people—more than 10% of Haiti's population—lived in emergency shelters across 1,555 camps (IASC E-Shelter and CCCM Cluster, 2012b).¹² Due to the threats of hurricane season and the extremely difficult living situations within the camps, informants felt pressure—oftentimes expressed in personal terms—to transition IDPs out of the *anba prelas*. Thus, even though some informants aimed to bypass the t-shelter modality, the majority said that due to the challenges of land tenure, the lack of space in Port-au-Prince and the living conditions within the camp, a t-shelter stage was necessary to ensure populations did not spend years living in tents. As of October 2012, more than 350,000 people remained in emergency shelters (IASC E-Shelter and CCCM Cluster, 2012b). Figure 2 is a picture of emergency shelters in the Autoparc Camp in Port-au-Prince.

Informants, often unprompted, brought up the utility of non-physical housing solutions, such as institutionally supported hosting, cash for rebuilding and rental subsidies. Though some agencies did have success with non-physical housing solutions, multiple informants noted that administrative and logistical challenges, particularly in a dense urban area, made implementation challenging. Examples of successful programming include funding provided by the US Agency for International Development (USAID) for hosting support to more than 26,000 households and the Rental Support Cash Grants provided to more than 14,000 families to allow them to move out of displacement camps (Setchell, 2011; Fitzgerald, 2012).

Figure 2. Emergency shelters in the Autoparc Camp, Port-au-Prince, 2010



Source: Joseph Ashmore.

Informants said that the magnitude and severity of the damage, as well as the media attention on the recovery process, had driven the sector towards more predictable and manageable mid-term shelter solutions, that is, t-shelters. As noted in the Emergency Market Mapping and Analysis (EMMA) report on corrugated galvanized iron, there was a preference for in-kind materials over cash among the displaced population. Local retail outlets were destroyed, which made purchasing difficult; furthermore, money was more likely to be stolen than materials (IRC et al., 2010).

Although t-shelters produced by different agencies were generally similar, minor design variations resulted in an increased need for different products (such as different sizes or specific types of timber), increased wastage and a decreased ability for joint purchasing. The shelter cluster tried to organise members to do joint purchasing and shipping. This, some informants felt, would have increased the speed of procurement, decreased the price of purchasing and shipping via increased purchasing power and limited total transport. However, practitioners did not take to the idea. The underlying reasons for the resistance to joint purchasing were a lack of congruence of shelter inputs, different internal regulations among implementing agencies and distribution concerns (such as the order in which the timber would be distributed and the resulting impact on agencies' operations). Informant 23, an implementer who was involved with sourcing, noted:

However big [the needs] looked on the ground, to anybody standing in the middle of Port-au-Prince, when you compare those needs to the normal North American construction industry, they become dwarfed. So part of the issue was that single NGOs would go up to Canada and go into all the major timber suppliers and would not be able to get their attention even if they submitted a procurement order. Because it just was not big enough to be cared about with much larger commercial orders coming through.

Timber procurement

Compared with responses to other mega-disasters, the one in Haiti saw timber sourced relatively sustainably. It is unclear whether the environmentally responsible procurement of timber was due to training or partnerships with environmental organisations, situational factors related to Haiti's geography and the timing of the earthquake—or rather increased ES awareness among disaster personnel. Informants demonstrated a sophisticated understanding of timber-related environmental issues and explained that they tried to—and were able to—procure certified sustainable or at least certified legal timber with certificates of origin. While donors and the shelter cluster encouraged sustainable sourcing, it was initiated entirely by the implementers and generally without external support. Different agencies cited various initiatives to ensure sustainable timber for their transitional shelters. These included:

- purchasing only Forest Stewardship Council (FSC) timber from the United States and Canada;
- procuring timber from an FSC-certified forest in the Dominican Republic; and
- using beetle-killed wood from the western United States.¹³

There has been a considerable effort to increase the use of sustainable timber in post-disaster settings since the Boxing Day tsunami. Amongst others, the World Wildlife Fund, the shelter cluster, and the International Federation of the Red Cross and Red Crescent Societies all issued timber guidelines, and many NGOs have timber policies that guide sustainable purchasing. In addition, technical documents—such as the Green Reconstruction and Recovery Toolkit (GRRT)¹⁴ and the Rapid Environmental Impact Assessment (REA)¹⁵—provide technical assistance on the procurement of sustainable timber. Furthermore, for much of the implementation phase, the shelter cluster had at least one environmental advisor who was able to advise on timber issues.

Haiti-specific factors related to location and geography also increased the likelihood of sustainable timber. What stands in sharp contrast to the recovery following the 2004 Boxing Day tsunami is that there were no reports of unsustainable timber coming from the immediate area, meaning the Dominican Republic; Haiti had insufficient timber sources to support the reconstruction. This made it easier to confidently identify sustainable and/or legal timber as there was limited, if any, mixing of supply chains.¹⁶ As noted by Informant 29, a donor with an environmental background:

[When compared to Haiti], a country like Indonesia is way more complex because there are internal supply chains of timber and [those are mixed with legal and illegal timber] and then you start dealing with a lot of [supply chain] issues.

Informants also noted that the US housing crisis resulted in a surplus of timber in the US states bordering on the Gulf of Mexico. A significant portion of the timber used in t-shelters was reportedly southern yellow pine, which is grown in the Gulf States. As a result, there was a surplus of quality, reasonably priced timber nearby. As noted by Informant 20, an implementer:

Because [the earthquake] hit during the lull in the housing market—where most of the lumber would be normally going—a bunch of these American lumberyards had a good inventory in Florida and all along the Gulf Coast. Three or four years earlier that might not have been the case.

Procurement of concrete

Concrete was used primarily as a slab to ensure that the t-shelters could withstand hurricane winds (IASC Haiti Shelter Cluster, 2010, p. 5).¹⁷ An estimated 80,000–120,000 m³ of sand and aggregate was used for the concrete in t-shelters (Navaratne, 2010, pp. 19–23). This volume may pale in comparison to the total used in long-term housing, as the t-shelters were predominately timber-framed. Informants who knew the source of the aggregate used in concrete formation noted that it was predominantly sand and gravel from rivers and mountains in Haiti—probably unsustainable and potentially hazardous sources.

Unsustainable procurement of aggregate can have multiple environmental consequences. Uncontrolled extraction of sand from mountains (as reportedly took place in Haiti) increases the likelihood of landslides and topsoil erosion, putting nearby communities at risk. Moreover, it can also result in contaminated runoff, which can affect water supplies. The removal of sand from a river can cause silting and damage to the river; as a result, the passage of light through water may become limited and basic organisms at the bottom of the food chain may be negatively affected. The deposited sediment caused by sand mining can destroy fish habitat and disrupt marine food chains (GRRT, 2010, module 5).

Multiple informants were aware of the consequences of unsustainable aggregate procurement. During an interview, one implementer pointed out a massive scar on a mountain and explained that mining for sand and aggregate was the cause (the scar was clearly visible even from a distance).

More salient than informants' understanding of the risks of unsustainable aggregate procurement was the perception that the challenges and trade-offs for sustainable procurement were too high. That is, even though t-shelter implementation is not an emergency activity and unsustainable procurement could have exacerbated localised risks, informants valued speed more than ES. As noted by Informant 16, an implementer:

I think everyone recognised that sand and gravel was probably coming to us in a very environmentally unfriendly way, but no one dug too deeply into it. Because it was such a scarce commodity that you got it from wherever you could get it.

Despite significant challenges to rubble management and reuse (such as the lack of infrastructure, high population density and administrative difficulties in coordination and removal), multiple agencies demonstrated that rubble-to-reconstruction programmes are feasible. One organisation used rubble as aggregate for concrete blocks and in concrete slabs; another organisation created a factory-type setting to crush the rubble and produce usable items for local communities (such as drain covers).

Consistent with those involved in t-shelter implementation, informants involved in rubble reuse programmes were motivated most by speed and the economic aspects of the recovery—to them ES was an ancillary benefit.

Although many informants expressed concern over using the rubble as aggregate in load-bearing structures, data reveals that there were plenty of possible non-load-bearing uses. For example, direct observation demonstrated that gabion blocks were used as retaining walls throughout the country.¹⁸ Yet, in some areas, pieces of foam were propped up near rivers as clearly inadequate retaining walls. In addition, rubble represents a unique opportunity to implement large-scale creative programmes. Informants mentioned that in other urban areas, rubble was reused as landfill or for public works programmes. For example, following the 2005 earthquake in Pakistan, rubble was reused to create a green space in Muzaffarabad.

Identified barriers to ES

Following the 2010 earthquake in Haiti, agencies' explicit incorporation of ES was sporadic and inconsistent. This is explained by interrelated barriers that stemmed from (1) prioritisations and perceptions within the disaster response sector, and (2) structural and organisational barriers within the disaster response framework.

ES depended on the motivation and capacity of specific actors and agencies, particularly implementers. Issues of prioritisation and perception within the disaster response community resulted in limited enthusiasm for incorporating ES into programming. Prioritisation- and perception-related barriers include:

- a perceived trade-off between speed and ES;
- a muddled delineation between emergency response and redevelopment; and
- the fact that non-environmental concerns drive decision-making.

The agencies that attempted to incorporate ES into programming faced specific structural and organisational barriers. Those barriers served as a further disincentive to incorporating ES into programming. They include:

- unclear accountability for ensuring ES;
- a lack of personnel with ES and disaster expertise;
- the humanitarian sector's slow rate of adaptation to new practices; and
- the logistical difficulty of pre-positioning and pre-planning.

Prioritisation and perception barriers

A perceived trade-off between speed and ES

Informants were adamant that if faced with a choice of ES or speed, ES would not be possible. Disasters come with an enormous pressure to be fast. Implementers attributed this pressure to the media, donors and a personal drive to help victims. As noted by Informant 20, an implementer:

It's the nature of emergencies. Everything goes out the window in terms of being environmentally or fiscally conscious. CNN is there showing people starving to death in the rain and no one is going to say, 'We can fly things in by plane but it has a higher carbon output than doing it by ship.' It's just the urgency of [the situation].

However, informant 31, who has an environmental and disaster background, provides a conflicting view:

I think the urgency of disaster response is an excuse more than anything not to do good programme design. I have seen quite a few organisations come in after a disaster and say, 'We can't do good environmental management because we have to act quickly.' And that's actually contradictory to what should be done. You don't have to take more time to take environmental issues into account when you are responding to a disaster. You just need to think through all the implications of what you are doing today and how they are going to affect people and social, economic and systemic factors tomorrow and the next day and the next year.

The drive for speed had a direct impact on material choice. Informants explained that they could not slow their process to look for sustainable sources, and that the short-term pressure outweighed more long-term concerns such as ES. Even if a material was known to be unsustainable and came with possible mid-term risks, the ability to procure it quickly was still the primary decision factor. As noted by Informant 16, an implementer:

I think part of the issue is that with a lot of these disaster-response programmes the focus is on high-speed implementation and hitting the numbers, so there is very little latitude to go back to a donor and say, 'Well, we are going to slow down because we know this gravel is coming from the local river.'

A muddled delineation between emergency response and redevelopment

According to some informants, there is an industry-wide failure to clearly delineate between recovery and re-development activities. This, informants argued, restricts proper planning and risk management (such as environmental analysis) and thus disaster resilience. Informants noted that in Haiti the lack of delineation between emergency and recovery or redevelopment efforts was pronounced. They attributed this to the severity of the earthquake and prior socioeconomic challenges. Moreover, due to the many earthquake-related challenges in the months that followed the initial impact (including the cholera outbreak), life-saving activities were taking place for far longer than expected. When asked, most informants had trouble distinguishing recovery from development activities. Consistent with the literature, informants described response efforts not as phases, but as overlapping steps, noting that redevelopment starts in the first few weeks, or even days, after the disaster.

Yet the emergency approach and priorities tend to persist in the long term. As a result, according to informants with environmental backgrounds, humanitarian actors

become accustomed to focusing on the urgency and speed of disaster response and never shift into a mindset focused on long-term planning and redevelopment—phases in which ES should be explicitly pursued. Therefore, ES is addressed too late, or not at all. As noted by Informant 30, an environmental expert:

The reconstruction [phase] is part of the development continuum [. . .]. If [we] are going to spend the next five years working on this, we need to approach it like it's a development project: proper planning and an environmental assessment is part of that process.

Non-environmental concerns as drivers of decision-making

Although interest in ES within the disaster-response community is growing, ES was not a primary concern in view of the multitude and immediacy of other challenges. Instead, speed, logistics and socioeconomic concerns drove decision-making.

The challenges of shipping to Haiti and navigating an overwhelmed customs agency, which some informants called corrupt, were a source of frustration, particularly for implementers. Many NGOs that tried to procure materials directly often reverted to working with Haitian intermediaries. Simply put, Haitian businesses were better at navigating customs than foreign NGOs. A donor explained that grantees tried different strategies for procurement; when factoring in delays in customs, procuring locally was more efficient and less expensive than trying to import directly. That remains true even when the source origin of the materials was the implementing agency's home country. Partly for that reason, informants reported that in Haiti, timber—most of which came from overseas—was purchased through local importers or retailers.¹⁹ As noted by Informant 28, a donor:

The partners that did go through local procurement [faced] less problems in customs than the ones who were shipping it directly from external [sources]. Because dealers in Haiti, they already knew how to navigate the system.

In addition to receiving necessary materials quickly, many informants felt that purchasing through Haitian intermediaries also helped stimulate the Haitian economy.²⁰ Some informants indicated that the economic effects of shelter were more important than the physical benefits. An expert in Haitian shelter noted that Haitians ranked shelter as the sixth greatest need, after livelihoods, education, water, food and health care. This was also reflected in the UN's 2011 annual report on Haiti operations, which notes: 'Displaced people living in camps, identified employment as being their first preoccupation, before other needs such as housing or access to education' (UN, 2011, p. 43).

Informants were aware of Haitians' need and desire for employment. They were adamant that they should—and did—do everything in their power to support the local economy and the employment of Haitians. As noted by Informant 16, an implementer:

We made the decision with the Haitian economy the way it was, unless there was significant savings in [purchasing directly], it was probably better to invest our funds locally so it

generated livelihood opportunities for local Haitians and put some money in the local economy. It would be a bit short-sighted [not to purchase locally], for a 5% price difference.

Informants explained that timber and cement were chosen, at least in part, because there are many masons and carpenters in Haiti, and therefore building with timber and cement was compatible with the skills of the Haitian people. The economic considerations also informed the selection of shelter structures. This ruled out any prefabricated, modular shelters built abroad. As noted by Informant 15, an implementer:

I would try not to do the prefabricated [shelters], just because I like to source commodities locally because you do get that cash injection into a local economy, which produces ripple effects and helps jumpstart businesses.

In addition to economic considerations, informants were conscious of the cultural appropriateness of the shelters and the shelter inputs. T-shelters could not be built in the abstract without considering the prospective inhabitants and their culture. As such, atypical structures (such as modular plastic housing) would not be used even if they were environmentally sustainable and less expensive. The heavy weighing of economic and cultural factors in Haiti reiterates a critical, overarching lesson: approaches to incorporating ES into recovery operations are more likely to be successful if they work with—or at least not against—goals related to economic and social sustainability.

Structural and organisational barriers to ES

Unclear accountability for ensuring ES

All parties expressed interest in implementing ES, yet there was confusion regarding who was responsible for the development, programming and funding of ES initiatives and for incorporating ES into programming (that is, should ES be funded explicitly or is it an expected cost of doing business?). Ultimately, it came down to the motivation and capacity of the implementers, many of whom were already overwhelmed by the magnitude of their operations.

Some of the leading implementers (measured in total t-shelters implemented) explicitly took ES into consideration. Specifically, they worked with environmental actors to ensure that materials came from non-damaging sources, that projects were sited properly and that their actions did not create further risks. It is of note that those were recommendations made in the REA, Sphere Standards and the GRRT.²¹ Informants found that with proper and early planning, there were limited challenges in implementing ES into their procurement and programming. As noted by Informant 13, an implementer whose agency explicitly considered ES:

[Considering ES did not affect our programming] at all. Because we had [our environmental partner] early on. It wasn't like we bought the products and then they told us it would be problematic. We went through the bidding process for the materials after we had many discussions with them about environmental concerns, so it wasn't a problem [. . .].

I would have remembered if [ES] had a big impact, it might have cost a bit more, but I don't remember it being a very marked difference in terms of cost.

Other implementers indicated that it was difficult to consider ES because donors did not provide the resources or motivation to pursue it. As noted by Informant 16, an implementer, regarding environmental assessments:

Obviously, if it was mandated, we would have taken the pause to do an [environmental] evaluation or review, but if it is not mandated, it's not going to happen. And it's not just us, I am sure that would be the case with all NGOs. Because there is such pressure to get going.

Donors explained that they did not make specific ES requirements because they felt they lacked sufficient on-the-ground knowledge to dictate environmental riders into requests for proposals and grants. They were cognisant of environmental issues, but there was a common, though not universal, concern that they could not stipulate specific environmental considerations without hindering implementers' capabilities. As noted by Informant 29, a donor, regarding incorporating ES into procurement:

Our t-shelters are built by [implementers]. We had no say in their procurement activities [. . .]. We did not put specific guidelines [that stated] this is a requirement that you need to follow. But it was promoted that you should go for [ES materials] as much as possible [. . .]. It was not a policy that was strictly enforced.

Yet, donor organisations do impose other limitations with which implementers were able to comply. For example, there were requirements designed to stimulate the local economy and to ensure that items were not procured from terrorist-supporting countries. Those systems and controls are analogous to what many environmental experts see as a donor-driven strategy to ensure ES, requiring analysis of the market to determine the best possible option and allowing waivers if they are needed or sensible.

Some donors and implementers maintained that ES was the responsibility of the shelter cluster. Although the shelter cluster continuously reminded implementers about the importance of ES, provided technical reports promulgating best practices, had on-site environmental advisors and held workshops on ES, it did not have enforcement capabilities. As noted by Informant 7, who worked for the shelter cluster, regarding timber procurement:

We always try and say to get the wood from a sustainable resource. But ultimately that is just a line in a document. People buy the wood from where they buy the wood and it is up to the agency how much effort they want to put into [ES].

A lack of personnel with ES and disaster expertise

Informants who had environmental as well as disaster experience pointed out that the challenge of incorporating ES into post-disaster recoveries had its roots in personnel

and staffing. Currently, there are few people and organisations with sufficient expertise in both disaster response and recovery and the environment. Thus, there are limited actors who can implement or oversee ES initiatives in post-disaster settings. This was reflected in the grey literature:

USAID/Haiti staff, normally responsible for environmental issues, are overworked and unable to dedicate sufficient time to the more than full-time requirements of overseeing a multi-million-dollar emergency programme (CHF International and Sun Mountain International, 2010, p. 22).

Donors and implementers explicitly noted the need for increased training, staffing and institutional support for ES. As noted by Informant 25, a donor:

I think it is very difficult in a post-disaster setting like that with the staffing that [we had] to actually walk over every single mitigation measure and recommendation of an environmental assessment. It's a matter of timing and staffing and workload.

The humanitarian sector's slow rate of adaptation to new practices

Asked why ES was difficult to implement, organisers, implementers and donors tended to cite that the humanitarian sector was slow to adapt to new practices. Kestle, Potangaroa and Storey (2008) note that the humanitarian community must constantly respond to new and different situations and, thus, there is no single model that can be applied to each recovery effort to ensure sustainability. Similarly, informants attributed the sector's slow rate of innovation to a pronounced need for reliability, a lack of funds and opportunities to field-test new techniques and technologies, and an inability to predict the location, timing and severity of disasters. Moreover, budgetary limitations result in general risk aversion. As noted by Informant 32, an implementer:

Innovation is happening at a very fast rate, and then it takes a while for it to trickle down into the humanitarian world and to have had enough road testing and be reliable enough and do what we need it to do.

The logistical difficulty of pre-positioning and pre-planning

According to Barakat (2003), pre-planning, as well as investments in pre-planning, can be a major factor in aiding ES in disaster response efforts; however, pre-planning is challenging due to the unpredictable nature of disasters. In donor-driven resource procurement, stockpiling and storing supplies—ostensibly obvious strategies for NGOs—creates other challenges (such as increased holding costs from warehousing and maintenance as well as obsolescence risks). Conversely, real-time procurement may carry a higher risk of a lack of preparedness and higher per-unit costs (Barakat, 2003). Informants directly weighed those trade-offs and concluded that, with the exception of select materials (such as plastic sheeting), pre-positioning only offers limited advantages. They argued that resource availability was a solvable problem and

that pre-positioning eliminates some of the economic impact that international actors could bring to the recovery. As explained by Informant 15, an implementer:

I think the move is away from pre-positioning relief supplies, and I know that sounds counterintuitive, but the reason why is there is increasing recognition of the resilience of markets in post-disaster settings and that recognition also basically factors in that whatever sourcing that can be done locally should be done locally. Because markets are so strong, so integrated that even with a [major disaster], markets will continue to function, and whatever local commodity you need, you can buy on the local market.

Conclusion

Working in a post-disaster setting is both complex and challenging, particularly in countries that have pre-existing socioeconomic, infrastructure and governmental challenges—such as Haiti. Post-disaster actors face an enormous challenge: balancing urgency and deliberation (Kates et al., 2007; Nelson, Ehrenfeucht and Laska, 2007; Olshansky, 2006). Integral to that balance is proper management of natural resources, that is, environmental sustainability (Kennedy et al., 2008; Smith and Wenger, 2007).

Incorporating ES into post-disaster activities is critical for risk mitigation and disaster resilience; not considering ES risks exacerbating or prolonging the damage of the disaster and hindering long-term socioeconomic goals.²² Utilising t-shelter implementation in Haiti following the January 2010 earthquake as a case study, this research has identified seven barriers to ES, which can be grouped into two categories: (1) prioritisations and perceptions within the disaster response framework, and (2) structural and organisational barriers within the disaster response sector.

Prioritisation and perception barriers within the disaster response framework include:

- a perceived trade-off between speed and ES;
- a muddled delineation between emergency response and redevelopment; and
- the fact that non-environmental concerns drive decision-making.

Structural and organisational barriers within the disaster response sector include:

- unclear accountability for ensuring ES;
- a lack of personnel with ES and disaster expertise;
- the humanitarian sector's slow rate of adaptation to new practices; and
- the logistical difficulty of pre-positioning and pre-planning.

This analysis of concrete and timber procurement reveals a sharp contrast in terms of ES: timber was reportedly procured predominantly from sustainable sources, while the aggregate used in concrete formation reportedly came mostly from unsustainable sources. More salient than the sustainability of the materials were the factors that drove purchasing. If ES was perceived to require trade-offs in speed, cost or local economic benefits—as was often the case—it would generally not be pursued.

Even if humanitarian actors wanted to pursue ES, structural, organisational and logistical barriers impeded implementation, providing further disincentives for incorporating ES into operations. As a result, ES came down to the capacity and motivation of the implementing agencies, many of which were already overwhelmed with the complex operations one would expect after such an extreme event.

The organisations that did not incorporate ES into their operations feared, perhaps incorrectly, that incorporating ES would be too costly in terms of time or money, or that the short-term challenges outweighed long-term concerns. Alternatively, these organisations were inhibited by a lack of knowledge about managing the environmental aspects of their programming. Actors were focused on the immediacy and urgency of the task at hand, and ES—an issue that is inherently long-term—was generally an afterthought. Yet informants who explicitly considered ES from the onset of their programming or who partnered with environmental actors were able to procure sustainable materials with limited, if any, trade-offs.

Although ES was a relatively low priority for disaster practitioners in Haiti, the interest in and recognition of its importance is increasing. Based on interviews conducted with humanitarian personnel regarding ES, Barrett, Murfitt and Venton (2007) find that despite some efforts to incorporate ES into the humanitarian response, it remained a low priority. This was attributed to limited environmental knowledge, high staff turnover and limited dissemination of best practices.

Informants interviewed for the current study echoed those concerns, but held that an understanding and appreciation of environmental issues was increasing. Informants provided specific examples of programming intended to increase the ES of their operations—for example, purchasing only certified sustainable wood, installing composting toilets and reusing rubble. In addition, informants noted the presence of environmental advisors and the shelter cluster's consistent reminders to incorporate ES into operations; moreover, one informant was working on developing environmental metrics for post-disaster shelter. Several informants discussed environmental interventions, often framed as programmatic tools; those solutions address some challenges of incorporating ES in a post-disaster setting.

As this research was a case study, not all its lessons will be applicable to other disasters. The lessons are most transferable to urban disasters in developing nations. This research focused on how the disaster response community can incorporate ES into their post-disaster operations. Future studies should examine how traditional environmental actors—such as environmental NGOs—can support post-disaster activities.

Acknowledgements

The author extends thanks to his advisors, Dr. Erica Schoenberger and Dr. Curt Ventriss, for providing invaluable guidance and advice, thoughtful questions and general support. He is also grateful to Anita van Breda of the World Wildlife Fund, who was a tremendous asset throughout the research process. He thanks the two reviewers for their insightful edits and comments and also acknowledges the informants

who graciously agreed to be interviewed. Special thanks go out to the two organisations that allowed the author to visit with them in Haiti and the people who were kind enough to guide him around—this research would not be complete without their generosity.

Correspondence

Daniel Abrahams, Department of Geography, University of South Carolina, Callcott Building, 709 Bull Street, Columbia, SC 29208, United States.

E-mail: abrahamad@email.sc.edu.

Endnotes

- ¹ The author was earning a Master in Public Policy at Johns Hopkins University while conducting research for this study, for which he received primary advisement from the University's Department of Geography and Environmental Engineering.
- ² Hazard and disaster have different meanings. A *hazard* is defined as: 'A dangerous phenomenon, substance, human activity or condition that may cause loss of life, injury or other health impacts, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage'; a *disaster* is defined as: 'A serious disruption of the functioning of a community or a society involving widespread human, material, economic or environmental losses and impacts, which exceeds the ability of the affected community or society to cope using its own resources' (UNISDR, n.d.).
- ³ Chang et al. (2010); Ingram et al. (2006); Kennedy et al. (2008); Nelson, Ehrenfeucht and Laska (2007); Olshansky (2006); Smith and Wenger (2007).
- ⁴ Chang et al. (2010); GRRT (2010); Guarnacci (2012); Ingram et al. (2006); Peacock, Dash and Zhang (2007).
- ⁵ GRRT (2010); Hendricks, Christensen and Toussaint (2010); Ingram et al. (2006); Mainka and McNeely (2011); Roseberry (2008); Srinivas and Nakagawa (2008).
- ⁶ Barrett, Murfitt and Venton (2007); GRRT (2010); Hendricks, Christensen and Toussaint (2010); Ingram et al. (2006); Mainka and McNeely (2011); Roseberry (2008); Smith and Wenger (2007); Srinivas and Nakagawa (2008).
- ⁷ See Goodland (1995) and WCED (1987).
- ⁸ Four of the informants were interviewed twice to fact-check unverifiable data and to provide clarity on specific issues. The total number of interviews was thus 41.
- ⁹ Informants routinely referenced land tenure (often unprompted) as well as obstacles related to security and poverty; however, the impact on ES was ambiguous.
- ¹⁰ The cluster approach is a strategy to address knowledge gaps and increase overall effectiveness through coordination and information sharing (OCHA, n.d.a). Or, as described by an informant, the cluster approach is a way to make sure all actors 'are reading from the same sheet of music'. Developed by the Inter-Agency Standing Committee, a coordination system within the United Nations, the cluster approach is activated based on four criteria: there is a trigger event (such as a disaster); an evaluation determines the country where the disaster took place is unable to respond adequately; there is a need for a multi-sectoral approach, which existing mechanisms cannot address; and the size of the operation warrants coordination efforts (OCHA, n.d.b). In Haiti, 69 partner organisations constituted the shelter cluster (IASC Haiti Shelter Cluster, 2010).

- ¹¹ Direct observation was consistent with Navaratne's (2010) presentation, which notes that nearly all t-shelters were timber-framed and most had concrete foundations or floors.
- ¹² According to the Emergency Shelter Cluster, '*Emergency shelter* consists primarily of tarpaulins and fixings, such as ropes, nails, a hammer etc. Tents can also be used for emergency shelter but, because they are less versatile than tarps, their use is limited. Emergency shelter can be distributed quickly but offers only limited protection against the heavy rains' (IFRC, UN-HABITAT and UNHCR, 2012, p. 16).
- ¹³ The use of beetle-killed wood not only provided less-expensive timber for Haiti operations, but also helped clear a US forest of a dangerous fire hazard, while at the same time decreasing total demand from traditional timber sources.
- ¹⁴ The GRRT is a training tool designed to 'provide information and guidelines to improve project outcomes for people and communities recovering from disaster by minimizing harm to the environment, and taking advantage of opportunities to improve the environment' (GRRT, 2010, module 3, p. 6).
- ¹⁵ The REA began roughly five weeks after the earthquake, on 16 February 2010, and concluded on 5 March. It identified potential environmental risks related to the earthquake and the recovery and provided recommendations to mitigate these risks.
- ¹⁶ Mixing is when legal or certified sustainable timber is mixed with illegal or undocumented sources of timber and purchasers cannot determine the source of timber.
- ¹⁷ Concrete is defined as the hardened combination of cement with aggregate and water. Cement is defined as 'a collective name for mineral binders in powder form, which set to become solid when mixed with water' (Berge, 2009).
- ¹⁸ Gabion blocks observed in Haiti were generally made with chicken wire cages and filled with rubble. A gabion house was made as a prototype t-shelter, though only a few were built.
- ¹⁹ It was reported that 90% of the timber came from North America; yet, informants from US-based NGOs reported purchasing through Haitian retailers (Potangaroa and Naidoo, 2010; Navaratne, 2010).
- ²⁰ A few informants questioned this theory, noting that there is limited data to support that purchasing through Haitian intermediaries helped the broader Haitian economy and not just a few individual Haitians.
- ²¹ The Sphere Project aims 'to improve the quality of their actions during disaster response and to be held accountable for them [. . .]. The Sphere Project identified minimum standards in key life-saving sectors which are now reflected in the Handbook's four technical chapters: water supply, sanitation and hygiene promotion; food security and nutrition; shelter, settlement and non-food items; and health action' (Sphere Project, 2004).
- ²² Barrett, Murfitt and Venton (2007); GRRT (2010); Ingram et al. (2006); Mainka and McNeely (2011); Roseberry (2008); Srinivas and Nakagawa (2008); van Breda and Laprade (2008).

References

- Alexander, D. (2004) 'Planning for Post-disaster Reconstruction.' Paper presented at the i-Rec 2004 International Conference on Improving Post-disaster Reconstruction in Developing Countries. Coventry, UK.
- Aquilino, M. (ed.) (2011) *Beyond Shelter: Architecture and Human Dignity*. New York: Metropolis, pp. 7–11.
- Ashmore, J. et al. (2003) 'Diversity and Adaptation of Shelters in Transitional Settlements for IDPs in Afghanistan'. *Disasters*. 27(4), pp. 273–87.
- Barakat, S. (2003) *Housing Reconstruction after Conflict and Disaster*. Network Paper Number 43. London: Humanitarian Practice Network, Overseas Development Institute.

- Barrett, E., S. Murfitt and P. Venton (2007) 'Mainstreaming the Environment into Humanitarian Response: An Exploration of Opportunities and Issues'. Prepared by Environmental Resources Management Limited for DFID.
- Berge, B. (2009) *The Ecology of Building Materials*. 2nd edn. Oxford: Architectural Press, ch. 13.
- Berke, P.R. and T. Beatley (1997) *After the Hurricane: Linking Recovery to Sustainable Development in the Caribbean*. Baltimore: Johns Hopkins University Press.
- Berke, P.R., J. Kartez and D. Wenger (1993) 'Recovery after Disaster: Achieving Sustainable Development, Mitigation and Equity'. *Disasters*. 17(2), pp. 93–109.
- Bernard, H. (2006) *Research Methods in Anthropology: Qualitative and Quantitative Approaches*. 4th edn. New York: Alta Mira Press, pp. 342–86.
- Chang, Y. et al. (2010) *Resources and Capacity: Lessons Learned from Post-disaster Reconstruction Resourcing in Indonesia, China and Australia*. Paper presented at the 2010 Construction, Building and Real Estate Research Conference of the Royal Institution of Chartered Surveyors, Paris. September. <http://unitec.researchbank.ac.nz/handle/10652/1565>.
- CHF International and Sun Mountain International (2010) *Rapid Environmental Impact Assessment: Haiti Earthquake—January 12, 2010*. http://pdf.usaid.gov/pdf_docs/PNADSO52.pdf.
- Corbin, J. and A. Strauss (2008) *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. 3rd edn. Thousand Oaks, CA: Sage Publications, ch. 7.
- Corsellis, T. and A. Vitale (2005) *Guidelines for the Transitional Settlement of Displaced Populations*. Oxford: Oxfam.
- Davis, I. (1978) *Shelter after Disaster*. Oxford: Oxford Polytechnic Press.
- FAO (Food and Agriculture Organization of the United Nations) (2007) *Good Practices for Hazard Risk Management in Agriculture: Summary Report Haiti—Project Phase I*. TCP/RLA/3101. <http://www.fao.org/docrep/010/a1501e/a1501e00.htm>.
- Fitzgerald, E. (2012) *Helping Families, Closing Camps: Using Rental Support Cash Grants and Other Housing Solutions to End Displacement in Camps*. Port-au-Prince: Inter-Agency Standing Committee Haiti Emergency Shelter and Camp Coordination Camp Management. http://www.eshelter-cccmhaiti.info/jl/pdf/Helping_Families_Closing_Camps2.pdf.
- Flyvbjerg, B. (2006) 'Five Misunderstandings about Case-Study Research'. *Qualitative Inquiry*. 12(2), pp. 219–45.
- Ganapati, N.E. and S. Ganapati (2009) 'Enabling Participatory Planning in Post-Disaster Contexts: A Case Study of World Bank's Housing Reconstruction in Turkey'. *Journal of the American Planning Association*. 75(1), pp. 41–59.
- Goodland, R. (1995) 'The Concept of Environmental Sustainability'. *Annual Review of Ecology and Systematics*. 26(1), pp. 1–24.
- GRRT (Green Recovery and Reconstruction Toolkit) (2010) *Green Recovery and Reconstruction: Training Toolkit for Humanitarian Aid*. San Francisco: World Wildlife Fund and American National Red Cross. http://green-recovery.org/?page_id=23.
- Guarnacci, U. (2012) 'Governance for Sustainable Reconstruction after Disasters: Lessons from Nias, Indonesia'. *Environmental Development*. 2, pp. 73–85.
- Hendricks, B., A. Christensen and R. Toussaint (2010) 'Green Reconstruction: Laying a Firm Foundation for Haiti's Recovery'. *Technology, Governance, Globalization*. 5(4), pp. 129–41.
- IASC E-Shelter and CCCM Cluster (Inter-Agency Standing Committee Haiti Emergency Shelter and Camp Coordination Camp Management Cluster) (2012a) 'Fact Sheet: Cluster E-Shelter and CCCM—May 2012'. http://www.eshelter-cccmhaiti.info/jl/images/pdf/factsheetmai2012_eng.pdf.
- IASC E-Shelter and CCCM Cluster (2012b) 'Fact Sheet: UCLBP and Shelter and CCCM Cluster Haiti—December 2012'. http://www.eshelter-cccmhaiti.info/jl/pdf/Cluster-CCCMShelter_120113_FINAL-eng.pdf.

- IASC Haiti Shelter Cluster (2010) 'Shelter Cluster Haiti: Transitional Shelter Technical Guidance'. 19 February.
- ICG (International Crisis Group) (2010) 'Haiti: Stabilisation and Reconstruction after the Quake'. Latin America/Caribbean Report No. 32. http://www.crisisgroup.org/~media/Files/latin-america/haiti/32_haiti___stabilisation_and_reconstruction_after_the_quake.ashx.
- IFRC, UN-HABITAT and UNHCR (International Federation of Red Cross and Red Crescent Societies, United Nations Human Settlements Programme, and United Nations High Commissioner for Refugees) (2012) *Shelter Projects 2010*. <http://www.sheltercasestudies.org/shelterprojects2010/ShelterProjects2010-lores.pdf>.
- Ingram, J. et al. (2006) 'Post-disaster Recovery Dilemmas: Challenges in Balancing Short-term and Long-term Needs for Vulnerability Reduction'. *Environmental Science and Policy*. 9(7–8), pp. 607–13.
- IPCC (Intergovernmental Panel On Climate Change) (2008) *Climate Change 2007: Impacts, Adaptation and Vulnerability*. Contribution to the Fourth Assessment Report of the IPCC Intergovernmental Panel on Climate Change. Geneva: IPCC Secretariat.
- IRC (International Rescue Committee) et al. (2010) *The Market System for Corrugated Galvanized Iron (CGI) Sheet in Haiti*. Port-au-Prince: IRC et al. <http://emma-toolkit.org/wp-content/uploads/2010/08/EMMA-Haiti-report-Corrugated-Iron-Market.pdf>.
- Joseph, M. and F. Wang (2010) 'Population Density Patterns in Port-au-Prince, Haiti: A Model of Latin American City'. *Cities*. 27(3), pp. 127–36.
- Kates, R. et al. (2007) 'Reconstruction of New Orleans after Hurricane Katrina: A Research Perspective'. *Cityscape: A Journal of Policy Development and Research*. 9(3), pp. 522.
- Kennedy, J. et al. (2008) 'The Meaning of "Build Back Better": Evidence from Post-Tsunami Aceh and Sri Lanka'. *Journal of Contingencies and Crisis Management*. 16 (1), pp. 24–36.
- Kestle, L., R.T. Potangaroa and B. Storey (2008) 'Towards the Validation of a Conceptual Design Management Model for Remote Site Projects?' *Proceedings of the 24th Annual Association of Researchers in Construction Management Conference*. Leicestershire, UK: University of Loughborough, pp. 185–94.
- Leon, E. et al. (2009) 'Capacity Building Lessons From a Decade of Transitional Settlement and Shelter'. *International Journal of Strategic Property Management*. 13, pp. 247–65.
- Lizarralde, G., C. Johnson and C. Davidson (2009) *Rebuilding after Disasters: From Emergency to Sustainability*. London: Spon Press, ch. 1.
- Mainka, S. and J. McNeely (2011) 'Ecosystem Considerations for Postdisaster Recovery: Lessons from China, Pakistan, and Elsewhere for Recovery Planning in Haiti'. *Ecology and Society*. 16 (1), art. 13.
- Navaratne, A. (2010) 'Environment and Shelter: Environmental Assessment of Haiti Shelter Projects'. PowerPoint presentation of the Inter-Agency Standing Committee Haiti Shelter Cluster. [https://www.sheltercluster.org/Americas/Haiti/HaitiEarthquake2010/Documents/Environmental Assessment Shelter Projects.ppt](https://www.sheltercluster.org/Americas/Haiti/HaitiEarthquake2010/Documents/Environmental%20Assessment%20Shelter%20Projects.ppt).
- Nazara, S. and B. Resosudarmo (2007) *Aceh-Nias Reconstruction and Rehabilitation: Progress and Challenges at the End of 2006*. Tokyo: Asian Development Bank Institute. <http://www.adbi.org/files/dp70.acehnias.reconstruction.rehabilitation.pdf>.
- Nelson, M., R. Ehrenfeucht and S. Laska (2007) 'Planning, Plans, and People: Professional Expertise, Local Knowledge, and Governmental Action in Post-Hurricane Katrina New Orleans'. *Cityscape: A Journal of Policy Development and Research*. 9(3), pp. 23–52.
- O'Brien, G. et al. (2006) 'Climate Change and Disaster Management'. *Disasters*. 30(1), pp. 64–80.
- OCHA (United Nations Office for the Coordination of Humanitarian Affairs) (n.d.a) 'Humanitarian Response Clusters'. <https://clusters.humanitarianresponse.info>.
- OCHA (n.d.b) 'Activate and Deactive Clusters'. <https://clusters.humanitarianresponse.info/activate-and-deactive-clusters>.

- Olshansky, R.B. (2006) 'Planning after Hurricane Katrina'. *Journal of the American Planning Association*. 72(2), pp. 147–53.
- OSE (Office of the Secretary-General's Special Adviser on Community-Based Medicine & Lessons from Haiti) (n.d.) 'Lessons from Haiti: Key Statistics'. <http://www.haitispecialenvoy.org/relief-and-recovery/key-statistics/> (accessed February 2013).
- Peacock, W.G., N. Dash and Y. Zhang (2007) 'Sheltering and Housing Recovery Following Disaster.' In H. Rodríguez, E.L. Quarantelli, R.R. Dynes (eds.) *Handbook of Disaster Research*. New York: Springer, pp. 258–74.
- Pelling, M. (2007) 'Urbanization and Disaster Risk'. Panel contribution to the Population–Environment Research Network Cyberseminar on Population and Natural Hazards, pp. 1–4.
- Potangaroa, R. and P. Naidoo (2012) 'TWiG: "T" Shelter Update'. PowerPoint presentation. <https://f6a2506e-a-62cb3a1a-s-sites.googlegroups.com/site/shelterhaiti2010/files/TShelterComparativeSurveypresentation.pdf>.
- Quarantelli, E. (1982) *Sheltering and Housing after Major Community Disasters: Case Studies and General Observations*. Columbus: Ohio State University.
- Roseberry, R. (2008) 'A Balancing Act: An Assessment of the Environmental Sustainability of Permanent Housing Constructed by the International Community in Post-disaster Aceh'. Paper presented at the i-Rec 2008 International Conference on Building Resilience: Achieving Effective Post-Disaster Reconstruction. Christchurch, New Zealand.
- Schoenberger, E. (1991) 'The Corporate Interview as a Research Method in Economic Geography'. *Professional Geographer*. 43(2), pp. 180–89.
- Setchell, C. (2011) *Hosting Support in Haiti: An Overlooked Humanitarian Shelter Solution*. Washington, DC: United States Agency for International Development. http://sheltercentre.org/sites/default/files/12.20.11_-_usaid-ofda-haiti-hosting-support-overview.pdf.
- Smith, G.P. and D. Wenger (2007) 'Sustainable Disaster Recovery: Operationalizing an Existing Agenda'. In H. Rodríguez, E.L. Quarantelli, R.R. Dynes (eds.) *Handbook of Disaster Research*. New York: Springer, pp. 234–57.
- Sphere Project (2004) *Humanitarian Charter and Minimum Standards in Disaster Response*. Geneva: Sphere Project.
- Srinivas, H. and Y. Nakagawa (2008) 'Environmental Implications for Disaster Preparedness: Lessons Learnt from the Indian Ocean Tsunami'. *Journal of Environmental Management*. 89(1), pp. 4–13.
- UN (United Nations) (2011) *Report of the United Nations in Haiti 2011: Rapò Nasyon Zini an Ayiti 2011*. http://www.onu-haiti.org/Report2011/UNHaiti_AR2011_Eng_Web1.pdf.
- UNEP (United Nations Environment Programme) (2011) *UNEP in Haiti 2010: Year in Review*. Nairobi: UNEP.
- UNISDR (United Nations International Strategy for Disaster Reduction) (n.d.) 'Terminology'. <http://www.unisdr.org/we/inform/terminology>.
- USAID (United States Agency for International Development) (2012) 'Fast Facts on the U.S. Government's Work in Haiti: Rubble'. <http://www.usaid.gov/where-we-work/latin-american-and-caribbean/haiti/rubble>.
- USCB (United States Census Bureau) (n.d.) 'State & County QuickFacts: New York County (Manhattan Borough), New York'. <http://quickfacts.census.gov/qfd/states/36/36061.html> (accessed February 2013).
- van Aalst, M. (2006) 'The Impacts of Climate Change on the Risk of Natural Disasters'. *Disasters*. 30(1), pp. 5–18.
- van Breda, A. and R. Laprade (2008) 'Reducing Risk and Vulnerability: An Environmental and Humanitarian Reconstruction Partnership'. *Proceedings of the 4th International i-Rec Conference*. TG 63: Disaster and the Built Environment, pp. 1–15. <http://www.irbnet.de/daten/iconda/CIB11581.pdf>.

- WCED (World Commission on Environment and Development) (1987) *Report of the World Commission on Environment and Development: Our Common Future* ('Brundtland Report'). Oxford: Oxford University Press. <http://www.un-documents.net/wced-ocf.htm>.
- Wisner, B. et al. (2004) *At Risk: Natural Hazards, People's Vulnerability, and Disasters*. 2nd edn. New York: Routledge.
- WWF and IASC (World Wildlife Fund and Inter-Agency Standing Committee Haiti Shelter Cluster) (2010) 'Timber Procurement Guidance Note for Haiti'. <https://www.sheltercluster.org/Americas/Haiti/HaitiEarthquake2010/Documents/Timber%20Procurement%20Guidance%20Note%20Haiti.pdf>.
- Zuo, K., and S. Wilkinson (2008) 'Supply Chain and Material Procurement for Post Disaster Construction: The Boxing Day Tsunami Reconstruction Experience in Aceh, Indonesia'. *CIB W89 International Conference on Building Education and Research BEAR 2008*. Sri Lanka: Heritage Kandalama.